

Skyscraper buildings with large windows could harness light for electricity. Researchers at the Los Alamos National Laboratory in New Mexico have reported that a thin film of quantum dots on glass could be the solution to achieving satisfactory efficiency in window photovoltaic systems at a low cost.

A thin layer of quantum dots on normal glass could have a lifetime of up to 14 years but low overall energy conversion efficiency of 1.9 per cent, which should be raised to 6 per cent for practical use. It's amazingly easy to add quantum dots to window glass as a machine pours slurry of quantum dots and polyvinylpyrrolidone polymer onto the glass and a blade spreads it out to form a thin sheet.

Quantum-dot solar cells may bring breakthrough innovation in the design of solar arrays with their favourable power-to-weight ratio and high efficiency. Mass and area savings as well as flexibility will result in miniaturisation, lower power consumption, increased efficiency, versatility and improved functionality of future satellites. The research also paves the way for new opportunities in the terrestrial renewable energy sector.

Quantum-dot solar window technology converts not just rooftops but complete buildings, including windows, into solar energy generators. It is envisaged that highly efficient and large-area colourless luminescent solar concentrators using heavy-metal-free colloidal quantum dots will soon be available in the market. Luminescent solar concentrators use a new sunlight harvesting technology that has the potential to turn any window into a daytime power source. Many academic research groups have shown that this technology holds promise. But very few companies such as Solterra Renewable Technologies in Texas claim to be competent in this area as it requires unique material chemistry and manufacturing know-how.

Solterra's cells are designed to exploit Quantum's low-cost synthesis methods for making four-armed quantum dots. The unique shape of these crystals reduces the probability of electron-hole recombination, which leads to greater charge transport and photovoltaic efficiencies than with spherical quantum dots of the same material.

Quantum-dot solar cells can be made more efficient but further progress in this area has been impeded by the challenge of understanding the mechanisms of electrical conductance in quantum-dot solids and the processes that limit the charge transport distance and low efficiency. The Los Alamos Lab of USA has recently solved this problem. This will lead to a dramatic boost in the photovoltage and overall device efficiency of quantum-dot solar cells. As economies of scale are achieved, quantum-dot solar cells will reach the market at competitive prices.

Perovskite: The most promising material

Days are not far off when solar cells will be sprayed or printed onto the windows of gigantic buildings, top of cars and walls at an unbelievably low cost. The main driver behind the low price of this technology will be

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM and LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT



Binay Opto Electronics Pvt. Ltd.

27, PANDITA TERRACE, Calcutta 700 029, India
Telephone: +91 (33) 4006 9875, 2475 0392, 2475 0030
email: info@binayled.com, http://www.binayled.com

POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983